Gene Cloning

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Introduction

Gene cloning is a technique to produce multiple copies of a specific gene. One method involves isolating the gene of interest (GOI), inserting it into a cloning vector, such as a plasmid, which is then inserted into a host organism, such as *E. coli*. This gene can then be replicated inside the host when it reproduces.

Below, is the GOI, which is to be isolated, as well as the vector into which it is to be inserted.

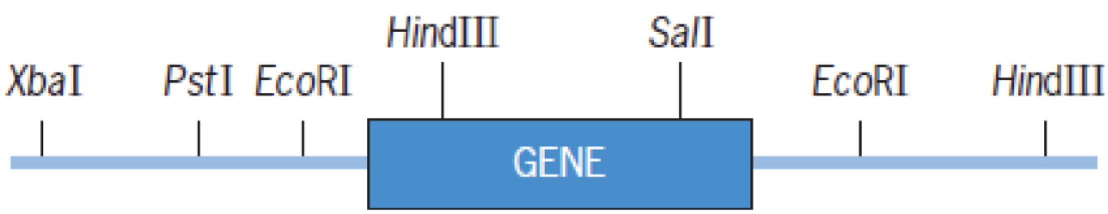


Figure 1: Gene of Interest

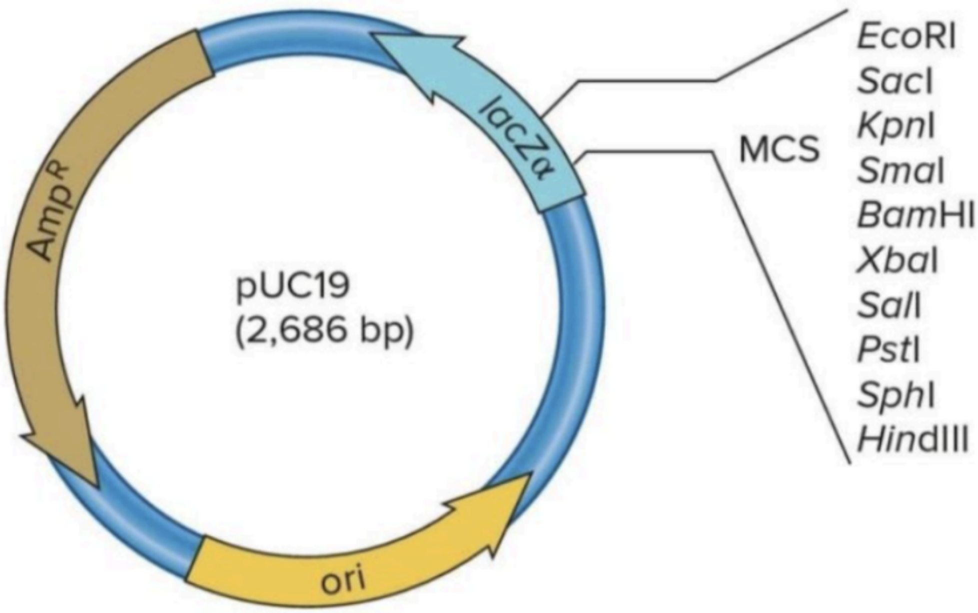


Figure 2: Plasmid Vector

Methods

1. We use the restriction enzyme EcoRI to cut out the GOI. This is because it has EcoRI restriction sites on both sides of it. All other enzyme sites either overlap with the gene of interest, or are present only on one side of the GOI.
2. Our plasmid vector (pUC19) has an Origin of Replication (ori), and two selectable markers: Ampicillin resistance (Amp^R), and the lacZα gene. The lacZα gene in the plasmid has multiple restriction sites, including the EcoRI site, within it This region is called multiple cloning site (MCS). This means, our GOI will be inserted within the MCS, inactivating the lacZα gene.
3. We treat both the GOI and the vector with EcoRI. To isolate the fragment containing the GOI, we use gel electrophoresis. This separates the fragments based on size, and allows us to isolate the gene, which is of a known size.
4. The GOI fragment and the vector are mixed, and a ligase is added. The complementary sticky ends base-pair via hydrogen bonding and the ligase seals the phosphodiester backbone of the sugar-phosphate chain, creating a circular recombinant plasmid.

5. Next, we must add these recombinant plasmids to organisms in which they can replicate. This process is called transformation. We use E. coli cells, since that is where the pUC19 plasmid is derived from. The strain used is DH5α competent cells. Competent cells are cells that have been treated to make them able to take up extracellular DNA.

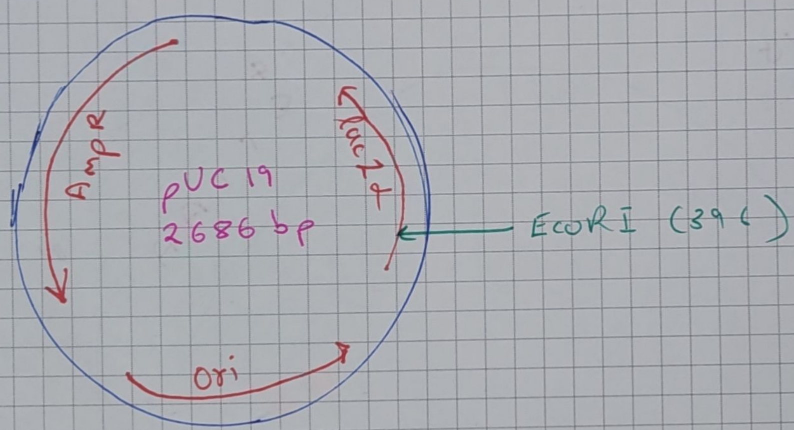
6. The cells are incubated, and left to grow in a medium containing the antibiotic ampicillin. Only the cells which have taken up the recombinant plasmid will survive. The cells which have not taken up the plasmid will die. This is how we make use of the Ampicillin resistance gene as a selectable marker.

7. To make use of the lacZα gene as a selectable marker, we also add IPTG and X-gal to the medium. X-gal is a chemical that is cleaved by the β-galactosidase enzyme. Upon cleavage, it turns blue. IPTG acts as an inducer, ensuring that the lacZα gene is expressed.

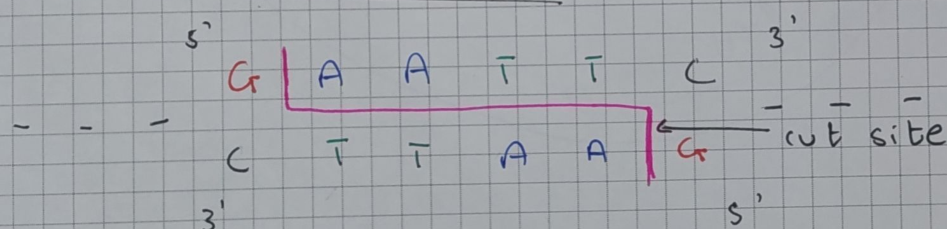
8. If the lacZα gene is intact, it will be expressed, and the cells will turn blue. This indicates that the GOI has not been inserted into the correct location, that is downstream of the lacZα gene's promoter. If the lacZα gene is disrupted by the insertion of the GOI, it will not be expressed, and the cells will remain white. We isolate the white colonies and grow them.

Note: Since we are using a single restriction enzyme to cut out both the GOI and the vector, there is a 50% chance that the gene will be inserted in the reverse direction, as shown in the image. However since this is a cloning vector and not an expression vector, the direction of insertion does not matter. The gene will be replicated regardless of its orientation.

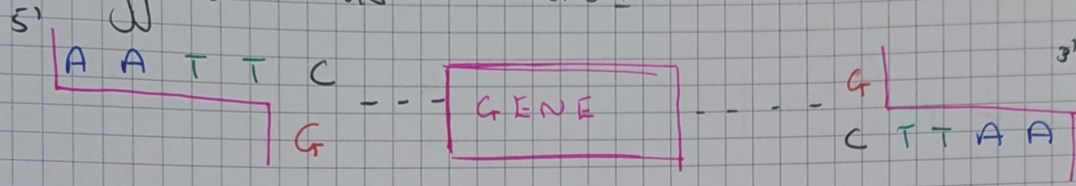
Plasmid Structure



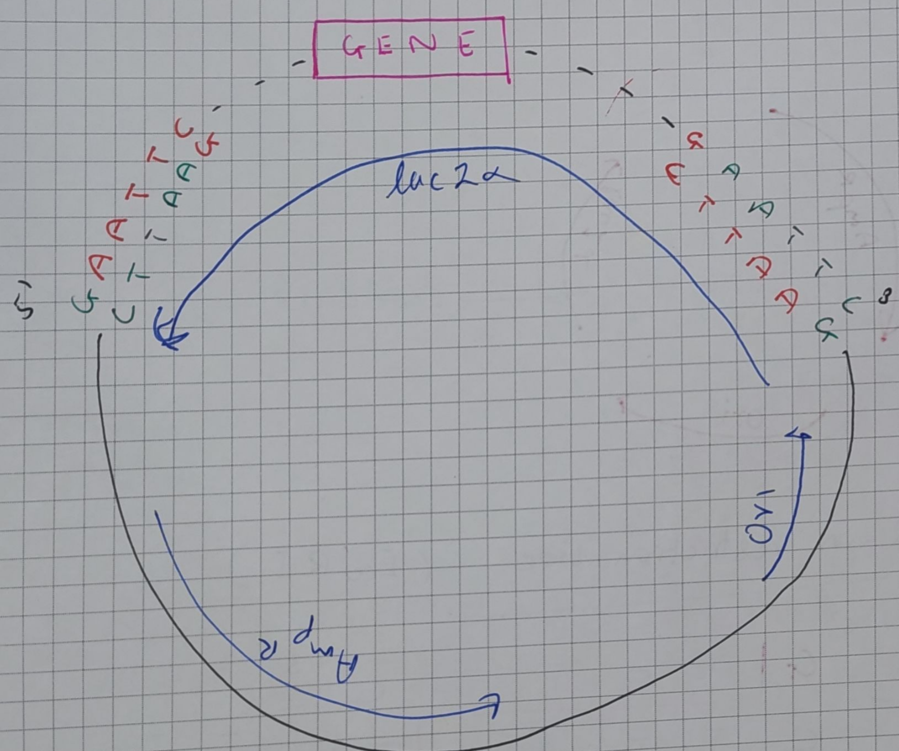
Zoomed-in restriction site : (EcoRI)



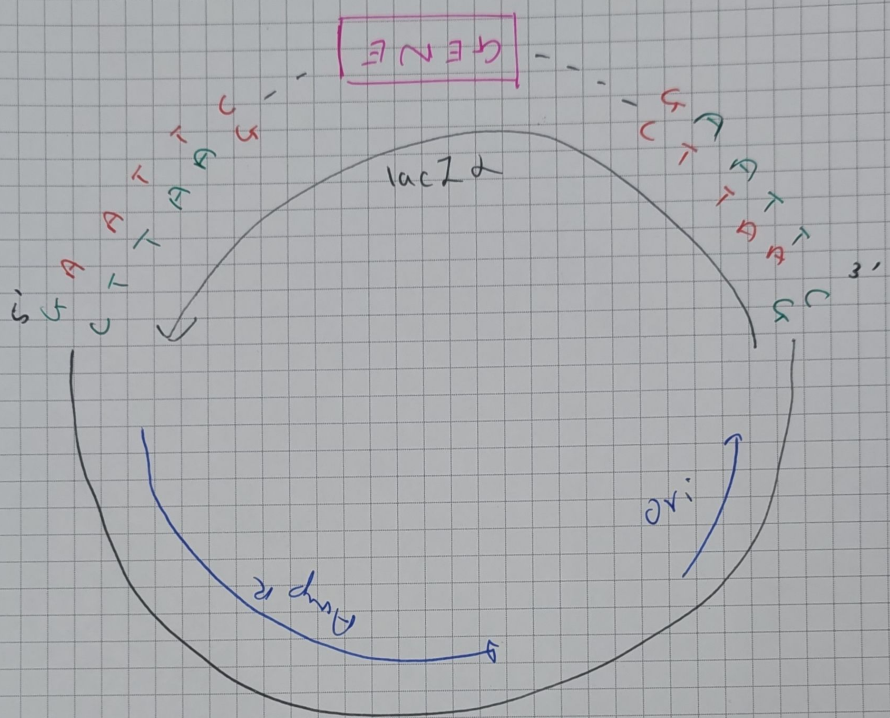
Staggered Ends in GOI :



Insertion into plasmid :



OR :



(inverted)

The diagrams are not drawn to scale

Green shows staggered ends of plasmid, red shows staggered end of gene of interest, in both diagrams